

SIMATS ENGINEERING

ASSESSMENT TOOL 2 (LOW) - Industry Problem Solving Task

COURSE NAME: R PROGRAMMING

COURSE CODE: ITA04

COURSE OUTCOME COVERED

CO3 Data Frames, Making data frames - Working with data frames- Data Reshaping- Melting and Casting of data – Merging Data Frames - Editing and Reading Data from Files –Reading and Writing Files.

(Bloom Level -BL4- BL5)

Assessment Tool Weightage: 25%

Code of Conduct:

*I **S. Sharon Surya (192324246)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sharon Surya

QUESTIONS:

Total Marks: 100

1. Healthcare Analytics

A hospital maintains two datasets: Patient.csv (PatientID, Name, Age, Disease) and Treatment.csv (PatientID, Doctor, TreatmentCost). Read both datasets and merge them to generate a report showing each patient's treatment details. Export the merged report as Hospital_Report.csv.

2. Retail Sales Analysis

A supermarket stores monthly sales data in separate CSV files for different branches. Read all datasets into R, merge them into a single data frame, and calculate the total sales for each branch. Save the consolidated report to a new CSV file.

3. Manufacturing Quality Control

A manufacturing company records product defects in separate production-line files. Combine the datasets, reshape the data to summarize defect counts by product category, and identify the production line with the highest defect rate.

4. Banking Customer Analysis

A bank maintains customer information in one file and loan details in another. Merge both datasets using the Customer ID and generate a report showing customers with loans exceeding ₹10,00,000. Save the filtered report.

5. E-Commerce Product Analysis

An online shopping company stores product details and customer reviews separately. Read both datasets, merge them, and reshape the data to calculate the average product rating for each product category.

6. Human Resource Management

An organization maintains Employee.csv and Salary.csv datasets. Merge the datasets, create a new column called Annual Salary, update missing salary values, and export the updated employee database.

7. Smart Agriculture Monitoring

A smart farming system records daily temperature, humidity, and crop yield. Reshape the dataset from wide format to long format using melt(), analyze environmental parameters, and reconstruct the original dataset using dcast().

8. Logistics and Supply Chain

A logistics company records shipment details in one dataset and delivery status in another. Merge both datasets to identify delayed deliveries and generate a city-wise shipment summary.

9. University Student Performance System

A university stores student details and semester marks in separate files. Read both datasets, merge them using Student ID, calculate each student's average marks, and export the final result sheet.

10. Insurance Claim Processing

An insurance company maintains policy details and claim records separately. Merge the datasets to identify customers with multiple claims, reshape the data to summarize claims by policy type, and save the processed report.

11. Telecommunications Network Monitoring

A telecom company records daily network performance metrics in multiple files. Combine all datasets, reshape the data to compare download speed across regions, and generate a performance report for management.

12. Energy Consumption Analysis

A smart electricity board collects hourly electricity consumption data from different regions. Convert the dataset from wide format to long format using `melt()`, summarize daily consumption using `dcast()`, and export the analysis.

13. Airline Reservation System

An airline maintains passenger booking details and flight schedules in separate datasets. Merge both datasets using Flight Number, identify delayed flights, update missing records, and save the final report.

14. Financial Market Analysis

A financial organization stores daily stock prices and company information separately. Read the datasets, merge them, reshape the stock prices to compare company-wise monthly performance, and export the summarized report.

15. Smart City Traffic Management

A smart city traffic monitoring system collects traffic volume from multiple intersections every hour. Read all traffic datasets, merge them, reshape the data to calculate peak-hour traffic by location, and generate a CSV report for urban planners.

1. Healthcare Analytics

```
patient <- read.csv("Patient.csv")
treatment <- read.csv("Treatment.csv")

report <- merge(patient, treatment, by = "PatientID")

write.csv(report, "Hospital_Report.csv", row.names = FALSE)

print(report)
```

```
> write.csv(treatment, "Treatment.csv", row.names = FALSE)
> patient <- read.csv("Patient.csv")
> treatment <- read.csv("Treatment.csv")
> report <- merge(patient, treatment, by = "PatientID")
> write.csv(report, "Hospital_Report.csv", row.names = FALSE)
> print(report)
```

	PatientID	Name	Age	Disease	Doctor	TreatmentCost
1	101	Rahul	35	Fever	Dr Kumar	5000
2	102	Priya	28	Diabetes	Dr Ravi	12000
3	103	Arun	40	Asthma	Dr Meena	8000
4	104	Kiran	50	Cancer	Dr Raj	25000
5	105	Meena	30	Migraine	Dr Anitha	6000

2. Retail Sales Analysis

```
b1 <- read.csv("Branch1.csv")
b2 <- read.csv("Branch2.csv")
b3 <- read.csv("Branch3.csv")

sales <- rbind(b1, b2, b3)

total_sales <- aggregate(Sales ~ Branch, data = sales, sum)

write.csv(total_sales, "Retail_Report.csv", row.names = FALSE)

print(total_sales)
```

```
<
> total_sales <- aggregate(Sales ~ Branch, data = sales, sum)
>
> print(total_sales)
```

	Branch	Sales
1	Chennai	290000
2	Coimbatore	303000
3	Madurai	301000

3. Manufacturing Quality Control

```
line1 <- read.csv("Line1.csv")
line2 <- read.csv("Line2.csv")

data <- rbind(line1, line2)

summary <- dcast(data, ProductCategory ~ ProductionLine,
  value.var = "Defects", fun.aggregate = sum)

print(summary)

highest <- aggregate(Defects ~ ProductionLine, data, sum)

print(highest)
```

```
> print(summary)
  ProductCategory Line1 Line2
1             AC      4      7
2           Fridge      1      3
3           Laptop      2      5
4           Mobile      5      6
5              TV      3      2
> highest <- aggregate(Defects ~ ProductionLine, data, sum)
> print(highest)
  ProductionLine Defects
1           Line1      15
2           Line2      23
```

4. Banking Customer Analysis

```
customer <- read.csv("Customer.csv")
loan <- read.csv("Loan.csv")

bank <- merge(customer, loan, by = "CustomerID")

result <- subset(bank, LoanAmount > 1000000)

write.csv(result, "Loan_Report.csv", row.names = FALSE)

print(result)
```

```
> write.csv(result, "Loan_Report.csv", row.names = FALSE)
> print(result)
```

	CustomerID	Name	City	LoanAmount
2	202	Priya	Bangalore	1500000
3	203	Raj	Hyderabad	1200000
5	205	Kumar	Delhi	2000000

5. E-Commerce Product Analysis

```
product <- read.csv("Product.csv")
review <- read.csv("Review.csv")
```

```
data <- merge(product, review, by = "ProductID")
```

```
rating <- aggregate(Rating ~ Category, data, mean)
```

```
print(rating)
```

```
> print(rating)
```

	Category	Rating
1	Accessories	5.0
2	Electronics	4.5
3	Furniture	3.5

6. Human Resource Management

```
employee <- read.csv("Employee.csv")
salary <- read.csv("Salary.csv")
```

```
emp <- merge(employee, salary, by = "EmployeeID", all.x = TRUE)
```

```
emp$Salary[is.na(emp$Salary)] <- mean(emp$Salary, na.rm = TRUE)
```

```
emp$AnnualSalary <- emp$Salary * 12
```

```
write.csv(emp, "Employee_Report.csv", row.names = FALSE)
```

```
print(emp)
```

	EmployeeID	EmployeeName	Department	Salary	AnnualSalary
1	1	Rahul	HR	30000	360000
2	2	Priya	IT	45000	540000
3	3	Arun	Finance	40000	480000
4	4	Kiran	Sales	50000	600000
5	5	Meena	Admin	35000	420000

7. Smart Agriculture Monitoring

```
farm <- read.csv("Farm.csv")
```

```
long <- melt(farm,  
            id.vars = "Day",  
            variable.name = "Parameter",  
            value.name = "Value")
```

```
print(long)
```

```
wide <- dcast(long, Day ~ Parameter, value.var = "Value")
```

```
print(wide)
```

```
7 Tue Humidity 58  
8 Wed Humidity 65  
9 Thu Humidity 55  
10 Fri Humidity 62  
11 Mon CropYield 200  
12 Tue CropYield 210  
13 Wed CropYield 205  
14 Thu CropYield 220  
15 Fri CropYield 215  
> wide <- dcast(long, Day ~ Parameter, value.var = "Value")  
> print(wide)
```

	Day	Temperature	Humidity	CropYield
1	Fri	30	62	215
2	Mon	30	60	200
3	Thu	32	55	220
4	Tue	31	58	210
5	Wed	29	65	205

8. Logistics and Supply Chain

```
shipment <- read.csv("Shipment.csv")
```

```
delivery <- read.csv("Delivery.csv")
```

```
data <- merge(shipment, delivery, by = "ShipmentID")
```

```
delay <- subset(data, Status == "Delayed")
```

```
summary <- aggregate(ShipmentID ~ City, data, length)
```

```
write.csv(summary, "Shipment_Report.csv", row.names = FALSE)
```

```
print(delay)
```

```
print(summary)
```



```

> print(delay)
  ShipmentID      City Status
2           2 Bangalore Delayed
4           4   Mumbai Delayed
> print(summary)
      City ShipmentID
1 Bangalore          1
2   Chennai          1
3     Delhi          1
4 Hyderabad          1
5     Mumbai          1
> |

```

9. University Student Performance System

```
student <- read.csv("Student.csv")
```

```
marks <- read.csv("Marks.csv")
```

```
data <- merge(student, marks, by = "StudentID")
```

```
data$Average <- rowMeans(data[, c("Sem1", "Sem2", "Sem3")])
```

```
write.csv(data, "Student_Result.csv", row.names = FALSE)
```

```
print(data)
```

```

> print(summary)
 StudentID  Name Department Sem1 Sem2 Sem3 Average
1           1  Amit         CSE   80   85   82 82.33333
2           2 Priya         IT   75   78   80 77.66667
3           3  Raj         ECE   90   92   88 90.00000
4           4 Divya        EEE   70   72   74 72.00000
5           5 Kumar        MECH   85   84   86 85.00000

```

10. Insurance Claim Processing

```
policy <- read.csv("Policy.csv")
```

```
claim <- read.csv("Claim.csv")
```

```
data <- merge(policy, claim, by = "PolicyID")
```

```
multi <- aggregate(ClaimID ~ CustomerID, data, length)
```

```
multi <- subset(multi, ClaimID > 1)
```

```
summary <- dcast(data,
  PolicyType ~ ClaimStatus,
  fun.aggregate = length)
```

```
write.csv(summary, "Insurance_Report.csv", row.names = FALSE)
```

```
print(multi)
print(summary)
```

```
> write.csv(summary, "Insurance_Report.csv", row.names = FALSE)
> print(multi)
  CustomerID ClaimID
1          1        3
> print(summary)
  PolicyType Approved Pending Rejected
1    Health         0         1         1
2     Life         2         0         0
3  Vehicle         1         0         0
```

11. Telecommunications Network Monitoring

```
n1 <- read.csv("Network1.csv")
```

```
n2 <- read.csv("Network2.csv")
```

```
network <- rbind(n1, n2)
```

```
long <- melt(network,
  id.vars = "Region",
  variable.name = "Metric",
  value.name = "Value")
```

```
report <- dcast(long,
  Region ~ Metric,
  value.var = "Value",
  fun.aggregate = mean)
```

```
write.csv(report, "Network_Report.csv", row.names = FALSE)
```

```
print(report)
```

```
> print(report)
  Region DownloadSpeed UploadSpeed
1 Central          96.5          39.0
2   East           91.0          35.5
3  North          102.5          41.0
4  South          122.5          51.0
5   West          112.5          46.0
```

12. Energy Consumption Analysis

```
energy <- read.csv("Energy.csv")
```

```
long <- melt(energy,  
  id.vars = "Region",  
  variable.name = "Hour",  
  value.name = "Consumption")
```

```
summary <- dcast(long,  
  Region ~ Hour,  
  value.var = "Consumption",  
  fun.aggregate = sum)
```

```
write.csv(summary, "Energy_Report.csv", row.names = FALSE)
```

```
print(summary)
```

```
> print(summary)  
   Region H1  H2  H3  H4  H5  
1 Central 110 115 120 125 130  
2   East   90  95 100 105 110  
3  North  100 120 110 130 140  
4  South  150 160 155 170 165  
5   West  130 135 140 145 150
```

13. Airline Reservation System

```
booking <- read.csv("Booking.csv")
```

```
flight <- read.csv("Flight.csv")
```

```
data <- merge(booking, flight,  
  by = "FlightNo",  
  all.x = TRUE)
```

```
delay <- subset(data, Status == "Delayed")
```

```
write.csv(delay, "Airline_Report.csv", row.names = FALSE)
```

```
print(delay)
```

```
> write.csv(delay, "Airline_Report.csv", row.names = FALSE)  
> print(delay)  
  FlightNo Passenger Status  
2    AI102     Priya Delayed  
4    AI104     Divya Delayed
```

14. Financial Market Analysis

```
stock <- read.csv("Stock.csv")
company <- read.csv("Company.csv")

data <- merge(stock, company, by = "CompanyID")
summary <- dcast(data,
  CompanyName ~ Month,
  value.var = "Price",
  fun.aggregate = mean)
write.csv(summary, "Stock_Report.csv", row.names = FALSE)
print(summary)
```

	CompanyName	Jan
1	HCL	1400
2	Infosys	1600
3	TCS	3500
4	TechM	1200
5	Wipro	500

15. Smart City Traffic Management

```
t1 <- read.csv("Traffic1.csv")
t2 <- read.csv("Traffic2.csv")
traffic <- rbind(t1, t2)
long <- melt(traffic,
  id.vars = "Location",
  variable.name = "Hour",
  value.name = "Traffic")
report <- dcast(long,
  Location ~ Hour,
  value.var = "Traffic",
  fun.aggregate = max)
write.csv(report, "Traffic_Report.csv", row.names = FALSE)
print(report)
```

```
> print(report)
  Location  H1  H2  H3  H4  H5
1 SignalA 125 155 185 165 145
2 SignalB 105 115 135 145 155
3 SignalC  95 105 125 135 115
4 SignalD 145 155 175 185 165
5 SignalE 115 125 145 155 135
```

RUBRICS FOR CALCULATION

Industry Problem Solving Task					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Application of EDA Techniques	25	Correctly applies central tendency, dispersion, and bivariate analysis techniques.	Applies most EDA techniques correctly with minor omissions.	Applies basic EDA techniques with limited accuracy.	Demonstrates limited or incorrect application of EDA techniques.
Statistical Analysis & Testing	25	Correctly performs and interprets T-test, ANOVA, and Chi-square analysis.	Performs statistical tests with minor interpretation errors.	Performs basic statistical analysis with limited interpretation.	Demonstrates poor understanding of statistical tests.
Problem Analysis & Data Interpretation	20	Accurately analyzes data patterns and derives meaningful insights.	Provides reasonable analysis with adequate interpretation.	Provides limited analysis and interpretation.	Analysis is weak or incorrect.
Decision-Making & Business Recommendations	20	Provides data-driven recommendations that effectively address the business problem.	Recommendations are relevant and supported by analysis.	Recommendations are basic with limited justification.	Recommendations are weak, unsupported, or missing.
Organization & Completeness	10	Response is well-structured, complete, and addresses all requirements.	Mostly complete and organized.	Partially complete with some missing components.	Incomplete and poorly organized.